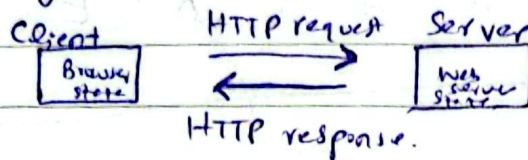


Small pieces of data that a web scraper sends to a client (browser)
Cookies
 Browser stores this data and send it back to the server.

- **HTTP (Hypertext Transfer Protocol)**
 Stateless (does not remember anything between requests)
 → Webpage opens + server responds (but it does not remember if we visit another page).

- Adding State



Web application maintains ephemeral (short-lived) state

- Use hidden fields
- Use Cookies

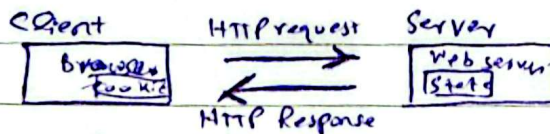
- Hidden Fields → includes an HTML form with hidden field + Session ID. in all the HTML pages sent to client. This hidden field will be returned back to server in the request.

Disadvantages:-

- Careful programming
- Session ends as soon as browser is closed.

Advantage:- All browser supports HTML forms.

- Statefulness with Cookies



- Secure Cookies

only sent to server with an encrypted request over HTTP protocol.

cookies track users.

Date: _____

- Visit a website → browser talks to website owner's server (first-party cookie) → Page may be connected to other companies' server (3rd party cookies) like ads → these can save third-party cookies that gives you an ID & tracks you across many websites.
↓
privacy reduces.

Examples- If we visit Apple website, we may see iPad ads on Amazon.

- If browser blocks third-party cookies, we can still be tracked using browser fingerprinting. Many small details can create a unique fingerprint like an ID.
• Small details = Browser & OS + Screen size + Language & time zone + Graphics type

- logged in to website → saves a session cookie → If attacker steals it, it can pretend to be you, access account without knowing the password.

Happen:-

- ① Site not using HTTPS
- ② Cookies are not secure
- ③ Public or unsafe wifi.

- Steal cookie ① Compromise server or user's machine
② Predict weak cookies

Network attack [③ Using HTTP instead of HTTPS
④ DNS poisoning]

- Defenses- Timeout session IDs & delete them once session ends
• Cookies should be random, long, impossible to guess.
• Use CSRF protections.